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CONTINUATION OF NOTES

Philip theuri

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1 PACKAGES, MACROS AND GRAPHICS

1.1 PACKAGES

There are different packages that are required in order for the compiler to run perfectly. They are usually included in the preamble that is between `\begin{document}` and `\documentclass`

- The backslash geometry is important as it helps customize every page as without it there will be a very large space left between the page label and the end of the page. you can however customize it to your own settings as done earlier.
- full page to change the format of the document ranging from width length to point of beginning of the document
- `amsmath` - Advanced mathematical equations; provides `align`, `cases`, `gather`, `split`, etc.-used to align two or more equations
- `amssymb` - Extra mathematical symbols such as $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$... special symbols like $\therefore$, $\because$, $\#$
- `amsmath` - Additional mathematical fonts; mainly used for commands like `\mathbb{b}`.
- `amsthm` - Creates theorem, lemma, corollary, proposition, and definition environments.
- `mathtools` - Extends `amsmath` with additional symbols and equation-formatting tools. `graphicx` - Inserts and resizes images.
- `float` - Controls the placement of figures and tables using options like `[H]`.
- `array` - Improves table and matrix formatting. especially where you need to place your figures center, left or right.
- `booktabs` - Produces professional-looking tables using `\toprule`, `\midrule`, and `\bottomrule`. these are much cleaner than repeated `\hline`

- multirow - Allows cells to span multiple rows in a table.
- geometry - Adjusts page margins and page dimensions.
- fancyhdr - Creates custom headers and footers. mostly used for assignments and reports
- hyperref - Makes references, citations, URLs, and the table of contents clickable.
- cleveref - Automatically formats references (Equation, Figure, Theorem, etc.).
- biblatex - Handles bibliographies and citations. manages references too...referring to einstein
- xcolor - Adds colours to text, tables, and other elements.
- listings - Displays source code with formatting.
- minted - Displays source code with advanced syntax highlighting.it is like writting a code
- tikz - Creates diagrams, graphs, flowcharts, vectors, and illustrations. eg tikzpicture or tikzcircuit
it is extremely powerful
- pgfplots - Draws mathematical graphs and plots.
- siunitx - Formats scientific units and measurements correctly. eg
- mhchem - Typesets chemical equations and formulas. eg
- multicol - Creates multiple columns of text.
- textcomp - Provides extra text symbols such as ° and €for additional symbols like the euro
- algorithm - Creates algorithm environments.
- algpseudocode - Writes pseudocode inside algorithm environments.
- enumitem - Customizes bullet and numbered lists.
- setspace - Controls line spacing (single, 1.5, double).
- caption - Customizes figure and table captions.
- subcaption - Creates subfigures and subtables.
- titlesec - Customizes section and chapter titles.

- appendix - Helps organize appendices.
- csquotes - Provides intelligent quotation marks and quotation environments.

NOTES

This is the symbol for the set of all real numbers; $\mathbb{R}$

This is the symbol for the set integers; $\mathbb{Z}$

This is the symbol of for the set if rational numbers; $\mathbb{Q}$

1.2 MACROS

A $\text{\LaTeX}$ macros is a user defined command that replaces a piece of text or code with a shorter command, allowing repetitive content to be written more efficiently and consistently throughout a document. It saves you time, make your document easy to maintain and avoid repeatedly typing the same code. it acts as a shortcut or abbreviation

example; $x \in \mathbb{R}$

for an equation; $y = \frac{x}{3x^2+x+1}$ we referred our equation as we defined it earlier at the preamble

1.3 GRAPHICS

in this case you use this command to insert images to you pdf. usually the package graphicx is used then for the command, type backslash include graphics for scale choose any betwee 0 and 1. then insert your image name say limit. you may or may not include the png or jnx format of the image file.